CSC 3320: System-Level Programming

Introduction to Course and UNIX

Spring 2026

Instructor: Dr. Md Mahfuzur Rahman
Department of Computer Science, GSU

Course Information

Instructor: Md. Mahfuzur Rahman, Ph.D.

Email: mrahman21@gsu.edu

Class Schedule: TR 12:45 PM – 2:00 PM

Location: Petit Science Center, Room: 169

Office Hours: TR: 11:00 AM – 12:30 PM or by appointment.

Office Location: 1 PP, Room: 637

Lab TAs and Schedule

Lab Schedule:

TA Name	TA Email	Lab Session	Location	CRN/Section
Hasan Mahmud	hmahmud1@student.gsu.edu	W:09:30 - 10:20	Classroom South 310	002
TBA	TBA	W:12:30 - 01:20	Classroom South 310	004
		W:11:00 - 11:50	Urban Life B. 302	006
		W:02:00 - 02:50	Classroom South 210	008

- Schedule applicable to 12:45 PM to 02:00 PM class (PSC 169)
- One Lab TA will be available during your lab session to help.
- Attendance is mandatory

LAB TA Office Hours

Lab TA Office Hours:

TA Name	Office Hours	Office Location
Hasan Mahmud	TBA	TBA
TBA	TBA	TBA

Ask the LAB TAs questions during LAB session for support.

Don't expect them to tell you the solutions during LABs

Expect elaboration and other technical support.

Meet them in-office hours for further help on class/lab materials

Graders

Grader Name	Email	Office	Office Hours
Shishir Kumar Vallapuneni	svallapuneni1@student.gsu.edu	TBA	TBA
Jeffrey Cris	jcris1@student.gsu.edu	TBA	TBA
Le Hoai Nhan Truong	ltruong18@student.gsu.edu	TBA	TBA
Sai Sarvagna Beeram	sbeeram1@student.gsu.edu	TBA	TBA

- Any of the graders can grade your work. If you have questions regarding your HW/Exam grades, keep all of them in the receiver list.
- Meet them in-office hours for further help on class materials

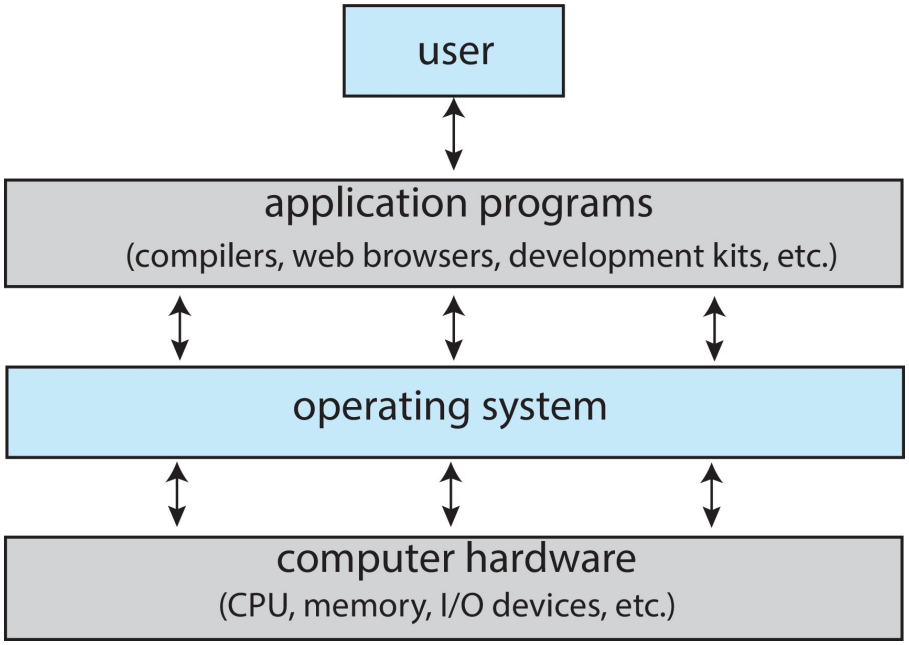
What is system programming?

- System (or system-level) programming is the practice of writing software that interfaces closely with the operating system and hardware—often via system calls—to manage core system resources and services, providing the foundational functionality on which application software depends.
  - managing system resources
  - performing I/O
  - file manipulation
  - process control and inter-process communication
  - network communication.

What to learn in this course?

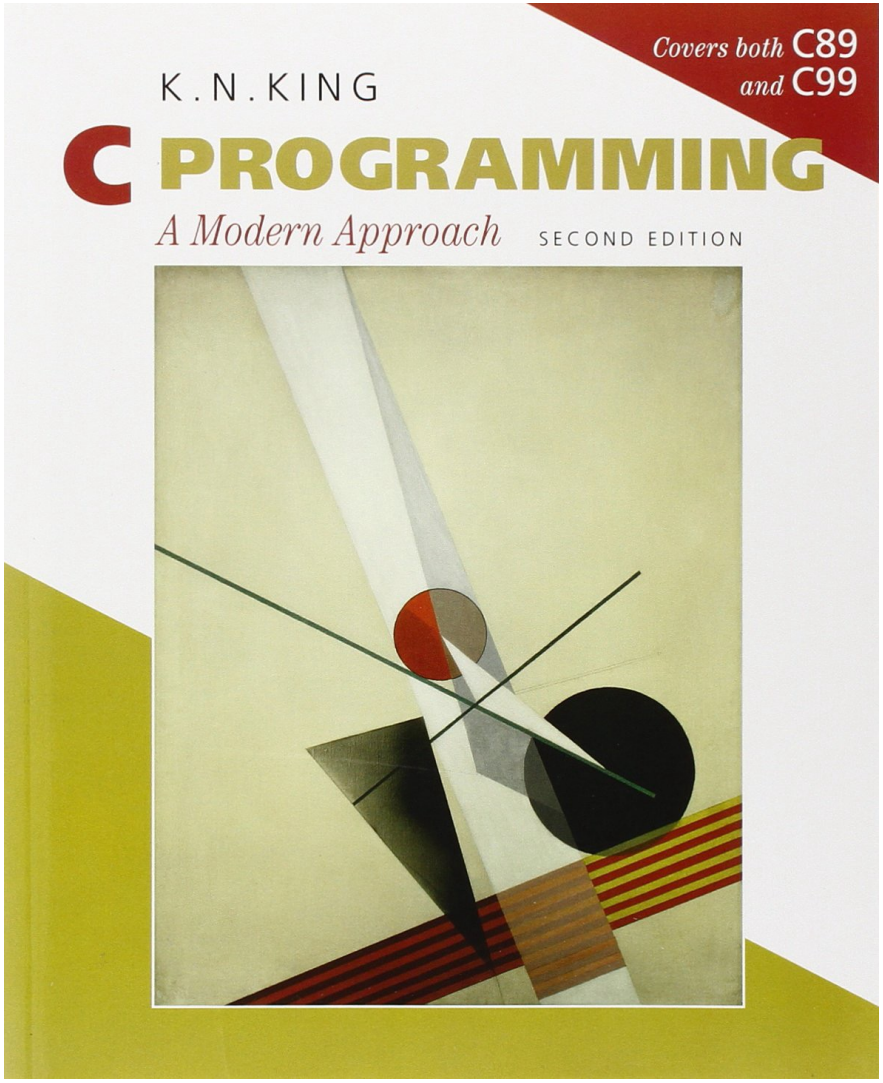
- It's a low-level programming course, so expect fundamental Unix concepts, shell scripting, and some C coding.

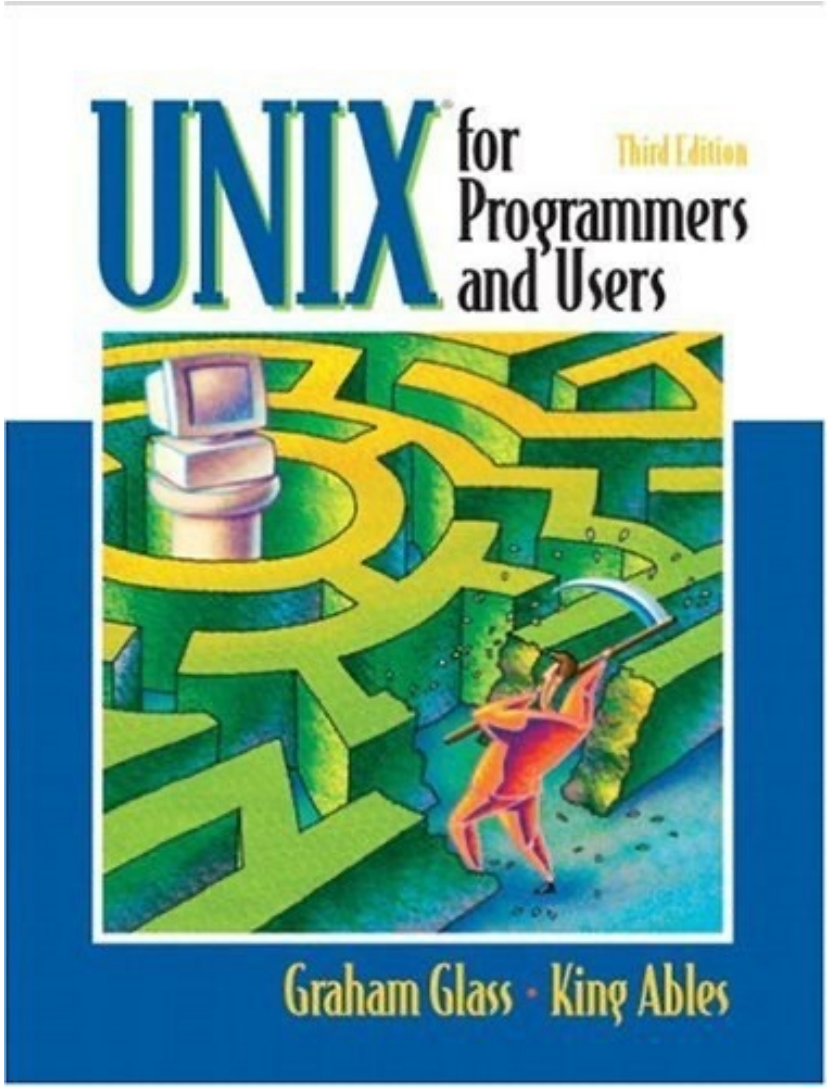
Why is system programming?



- SLP involves creating programs that manage or control hardware resources, such as memory, CPU, disk, or peripherals. Example:
- writing a driver software, OS, or creating utilities like file manager, scheduler, etc.
- provides foundational services to other software.

Textbooks


Covers both C89 and C99
K.N.KING
C PROGRAMMING
A Modern Approach
SECOND EDITION


UNIX for Programmers and Users
Third Edition
Graham Glass · King Ables

Prerequisites

- (CSC 1302 or or DSCI 1302) and (CSC 2510 or MATH 2420) with a of C or higher.
- If you do not meet these requirements, please drop.

Grading Policy

- **Homework** (approx. 5): 15% (lowest score will be dropped)
- **In-class quizzes/participation:** 10% (two will be dropped, no make-ups!!)
- **Two exams:** 20% each
- **Lab Assignments** (approx. 14): 15% (lowest score will be dropped)
- **Final Exam:** 20%
- Graduate students (Master's or above) will have project instead of **Final Exam** (if there is any)

Course Policy

- Homework and LABs must be turned in **iCollege**
- We will use gcc compiler and **snowball.cs.gsu.edu** platform for our LABs.
- LAB must be submitted by the end (11:59 pm) of the following day after your lab session. (late submission not permitted for labs)
- For homework, 10% per day will be deducted from your score. Homework will not be graded after a week.
- Make-Ups: No make-up examinations for any unexcused absence. No make-up for the final exam and in-class quizzes.

Course Policy (contd.)

- **Class participation:** Top Hat will be used. Detail instructions will be given in **iCollege**
- Any gradable work must be your own.
- No use of Generative AI unless allowed in writing.
- No sharing of course materials on any website
- Communication: Use my email, not the iCollege messaging. Use “CSC3320…” in the subject line.
- Read carefully the entire syllabus.

How can you succeed in this course?

- Listen to the lectures carefully and take notes.
- Study the lecture materials thoroughly.
- Do not memorize things. Grow your knowledge.
- Start doing your homework earlier and submit timely.
- Show your uniqueness in your submission. No plagiarism please.
- You may be allowed to keep a 1-page paper (cheat sheet) to help you take some notes with you.
- The exams will be concept-based, will only test your knowledge.
- Above all: TRY NOT TO MISS ANYTHING (homework, exams, labs, in-class quizzes, etc.)

TOP HAT

- We will use the Top Hat platform for an engaging experience.
- New Registration: https://app.tophat.com/register/student
  - (using your school issued email address on web browser)
- Join Code:
  - Course Link: https://app.tophat.com/e/374686/
    Section join code: 374686 (12:45 to 2:00 Class)
- LAB attendance will be taken by Lab TAs.

Let’s start course topics

Expectations

- Knowing how to use Unix utility
- Writing basic shell script
- Learning C program
- System programming: File management
- System programming: Process management

Why learn UNIX?

Free

Open-source

Secure

Well-specified/well-documented

VS

Check-in with students

Which language you already learned?

A) Python

B) Java

C) C or C++

D) Python and Java

E) Python, Java, and C/C++

F) Java, and C/C++

A python example

```
# Example of file manipulation in Python
file = open("testfile.txt","w")
file.write("Hello World")
file.close()
```

Python has standard libraries for system level programming
Example: os, subprocess, etc.

Another Python Example

```python
import os

# List all processes running on the system (Linux/macOS example)
def list_processes():
    print("Listing processes:")
    os.system("ps aux")  # `ps` command to show all processes


if __name__ == "__main__":
    print("System-level programming example in Python:")
    list_processes()
```

Why C, Not Python?

C:

- fundamental in developing operating systems
- It provides direct access to system resources.
- UNIX and parts of Microsoft Windows are predominantly written in C.
- C can manipulate hardware resources.
- Developers use C to write firmware, device drivers, and other critical software.
- Control and speed is significantly higher.

Python:

- Faster development times
- Easier to use for non-performance-critical applications.

Connecting to a remote server

- Snowball is an instructional server for the GSU Computer Science Department.
  - Used for teaching C Programming and UNIX/Linux computer architecture classes
  - IT uses the university single sign-on system.
  - Student submitted C-Programs will be graded on Snowball.
- Snowball Server Knowledge Guide
  - https://cscit.cs.gsu.edu/sp/guide/snowball

SSH Protocol

- The **Secure Shell Protocol (SSH)** is a network protocol for sending encrypted data over an unsecured network.
  - Used primarily for remote login and command-line execution.
- SSH utility consists of a **SSH client** and **SSH server**.
- Designed for Unix-like operating systems.

SSH Client

- UNIX-like operating systems such as Linux and macOS have both the SSH client and server built in.
- Windows is not based on UNIX and therefore does not support it natively*.

- For Linux and macOS users, SSH is utility in your respective Terminals.
- On Windows, download and use the Putty program.
  - *Windows PowerShell now implements a Bash Terminal so the SSH utility is available natively for all windows users since Windows 7

Login to SNOWBALL via SSH

- Open Putty, PowerShell, or Terminal on your PC.
- Type in ssh username@snowball.cs.gsu.edu
  - Replace username with your GSU Campus ID
- If this is your first time connecting, you may be asked to remember the key fingerprint.
  - Type yes then press ***Enter***
- Type in your GSU Password, then press ***Enter***

```
Last login: Wed Aug  7 23:16:17 on ttys001
(base) Mds-MacBook-Pro:~ mdmahfuzurrahman$ ssh snowball.cs.gsu.edu
The authenticity of host 'snowball.cs.gsu.edu (131.96.155.217)' can't be established.
ECDSA key fingerprint is SHA256:Rir34vs3KIWl5Nhu2Y0B2QaSq6shzVrxQ/AQ/veW1NY.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added 'snowball.cs.gsu.edu,131.96.155.217' (ECDSA) to the list of known hosts.
mrahman21@snowball.cs.gsu.edu's password:
Creating home directory for mrahman21.
   +
   |   GSU Computer Science
   |   Instructional Server
   |   SNOWBALL.cs.gsu.edu
   +
[mrahman21@gsuad.gsu.edu@snowball ~]$ ls
```

Use sftp to upload files

```
(base) Mds-MacBook-Pro:CSC 3320 System Programming mdmahfuzurrahman$ sftp snowball.cs.gsu.edu
mrahman21@snowball.cs.gsu.edu's password:
Connected to snowball.cs.gsu.edu.
sftp> put unix.txt
Uploading unix.txt to /home/mrahman21/unix.txt
unix.txt                                                    100%  188     7.5KB/s   00:00
sftp> put sysprog.txt
Uploading sysprog.txt to /home/mrahman21/sysprog.txt
sysprog.txt                                                 100%   84     3.4KB/s   00:00
sftp>
```

Now, check those uploaded files in SNOWBALL server

```
[mrahman21@gsuad.gsu.edu@snowball ~]$ ls
sysprog.txt  unix.txt
[mrahman21@gsuad.gsu.edu@snowball ~]$
```

For more on sftp, visit this site: https://linuxize.com/post/how-to-use-linux-sftp-command-to-transfer-files/

Downloading files using sftp

- To download a file, use the following command:

```
sftp> get filename
```

- To download a directory, use the following command:

```
sftp> get -r remote_directory
```

Questions?